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PROJECT REPORT

LLM Decoding Parameter & Sampling Analysis

Evaluating Temperature, Top-K and Top-P in Local Large Language Models

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& Sampling Analysis

Evaluation of Temperature, Top-K and Top-P in Local Language Models

Project detail	Description
Models	Llama 3.1 and Qwen 2.5 through Ollama
Benchmark	Five factual, reasoning and general prompts
Experimental design	Four parameter configurations; three generations per prompt/configuration/model
Implementation	Python, Pandas, NumPy, scikit-learn and Streamlit
Results source	Uploaded experiment_results.csv (40 experiment rows summarizing 120 generation attempts)

Abstract
This project investigates how decoding parameters affect the outputs of local Large Language Models (LLMs), using Llama 3.1 and Qwen 2.5 through Ollama. The benchmark contains five prompts: two factual questions, two reasoning questions and one general creative prompt. Four decoding configurations were tested, with three generations for each prompt, configuration and model. The uploaded results file contains 40 experiment rows, each combining three outputs, representing 120 planned generation attempts. The implementation calculates lexical diversity as one minus the mean pairwise TF-IDF cosine similarity and marks a prompt/configuration as deterministic when all three response strings are exactly identical. In the recorded data, Llama 3.1 had lower average lexical diversity and higher exact-match repeatability overall than Qwen 2.5. For both models, the highest-temperature/broadest-sampling configuration generally produced more lexical variation than the lowest configuration, although this pattern was not perfectly monotonic for every model and prompt. Factual accuracy is not automatically calculated by the current script; it requires a separate reference-answer or human-scoring procedure.

1. Introduction
Large Language Models generate text by selecting tokens based on probability distributions. Decoding settings influence which tokens are considered and how strongly high-probability choices are favored. These settings can affect response repeatability, wording and output variation. This project compares two locally served models using a small shared benchmark and a Python experiment pipeline.

2. Problem Statement
Different decoding configurations may change how consistent and diverse LLM responses are, and may also affect factual reliability. The project runs repeated generations under selected configurations and analyzes the collected outputs to explore these effects.

3. Objectives
- Compare Llama 3.1 and Qwen 2.5 using the same benchmark prompts.
- Test four combinations of Temperature, Top-K and Top-P.
- Collect three responses per prompt and configuration.
- Estimate lexical diversity using TF-IDF cosine similarity.
- Measure exact-match repeatability across repeated outputs.
- Provide a Streamlit interface and save batch results to CSV.
- Interpret findings while clearly identifying metric and design limitations.

4. Background

4.1 Temperature
Temperature controls the sharpness of the token-probability distribution used during sampling. Lower values generally favor higher-probability tokens more strongly and can increase repeatability; higher values can permit more variation. Actual behavior depends on the model and runtime.

4.2 Top-K

Top-K restricts sampling to the K highest-ranked candidate tokens. Smaller values narrow the candidate pool, while larger values allow more candidates.

4.3 Top-P

Top-P, or nucleus sampling, selects from a set of likely tokens whose cumulative probability reaches the selected threshold. Lower values typically restrict the candidate set; values closer to 1.0 allow a broader set.

4.4 Evaluation concepts

Consistency is repeatability across runs; lexical diversity is variation in wording; factual accuracy is whether claims are correct. These concepts are distinct: a response can be consistently wrong, and a more diverse response is not necessarily better.

5. Tools and Libraries

Tool/library	Role
Ollama	Runs local models and accepts generation options.
Python	Experiment automation and application logic.
Pandas	Reads benchmark data and saves CSV results.
NumPy	Numerical operations for metrics.
scikit-learn	TF-IDF vectorization and cosine similarity.
Streamlit	Interactive interface for live generation and saved results.
JSON	Stores the benchmark prompts.

6. Models and Benchmark

The code uses the Ollama model identifiers llama3.1 and qwen2.5. Exact model tags and versions should be included in the repository README for reproducibility.

ID	Category	Prompt
P1	Factual	What is the capital city of Australia?
P2	Factual	Who wrote the novel 'Pride and Prejudice'?
P3	Reasoning	If all bloops are bleeps, and all bleeps are blips, are all bloops guaranteed to be blips? Explain step by step.
P4	Reasoning	A bat and a ball together cost $1.10. The bat costs $1.00 more than the ball. How much does the ball cost?
P5	General	Write a short, inspiring 3-sentence poem about a rising sun.

The benchmark is intentionally small and demonstrates the pipeline, but it does not represent the full range of LLM tasks.

7. Experimental Methodology

7.1 Parameter configurations

Config	Temperature	Top-K	Top-P	Description
C1	0.0	10	0.5	Lowest temperature and narrowest candidate settings in this test
C2	0.3	40	0.9	Conservative/moderate setting
C3	0.8	40	0.9	Higher temperature
C4	1.2	100	1.0	Highest temperature and broadest candidate settings

Important design note: more than one parameter changes between C1–C4. Therefore, this experiment compares combined configurations; it cannot isolate the individual causal effect of Temperature, Top-K or Top-P. A future controlled experiment should change one parameter at a time.

7.2 Procedure and experiment size

- Load five prompts from benchmark_prompts.json.
- For each of two models, run every prompt under all four configurations.
- Generate three responses for each model–prompt–configuration combination.
- Calculate lexical diversity and exact-match repeatability for the three responses.
- Save the summarized results to experiment_results.csv.

The planned total is 2 models × 5 prompts × 4 configurations × 3 runs = 120 generation attempts. The supplied CSV contains 40 rows, one for each model/prompt/configuration combination, with the three outputs concatenated in runs_output.

8. Evaluation Metrics

8.1 Lexical diversity

The code applies TF-IDF vectorization to the three responses, calculates pairwise cosine similarity and defines lexical diversity as:

$$\textbf{Lexical diversity} = 1 - \textbf{mean(pairwise cosine similarity)}$$

Higher values indicate less lexical overlap. This is a wording-based measure, not a measure of factual correctness or semantic quality. Tiny negative values around zero in the CSV are floating-point artifacts and should be interpreted as zero.

8.2 Exact-match repeatability

The is_deterministic field is true only when all three response strings are exactly identical. This is a strict metric: equivalent answers with different wording count as non-identical. Exact matching does not establish factual correctness.

8.3 Factual accuracy

The supplied script does not compute a factual-accuracy score. The factual answers can be checked against reference answers, and the reasoning prompts can be scored using a written rubric. Any accuracy

claims in this report are therefore limited to qualitative inspection, not a comprehensive automated accuracy metric.

9. Results

The following results were calculated from the uploaded experiment_results.csv. Mean lexical diversity is averaged over the five prompts within each model/configuration. Exact-match rate is the percentage of those five prompt/configuration rows where all three outputs were identical.

Model	Config	Temp.	Top-K	Top-P	Mean diversity	Exact-match rate
llama3.1	C1	0.0	10	0.5	0.017	60% (3/5)
llama3.1	C2	0.3	40	0.9	0.165	40% (2/5)
llama3.1	C3	0.8	40	0.9	0.233	40% (2/5)
llama3.1	C4	1.2	100	1.0	0.395	20% (1/5)
qwen2.5	C1	0.0	10	0.5	0.118	40% (2/5)
qwen2.5	C2	0.3	40	0.9	0.268	20% (1/5)
qwen2.5	C3	0.8	40	0.9	0.415	0% (0/5)
qwen2.5	C4	1.2	100	1.0	0.383	0% (0/5)

9.1 Overall model comparison

Model	Mean lexical diversity	Exact-match repeatability	Data coverage
llama3.1	0.202	40.0% (8/20)	20 model/prompt/configuration rows
qwen2.5	0.296	15.0% (3/20)	20 model/prompt/configuration rows

Across all 20 rows per model, Llama 3.1's mean lexical-diversity score was 0.202 and its exact-match rate was 40% (8/20). Qwen 2.5's mean lexical-diversity score was 0.296 and its exact-match rate was 15% (3/20). In this benchmark, Qwen 2.5 outputs were more lexically varied on average, while Llama 3.1 more often produced exactly matching responses. These results describe this small benchmark and these specific configurations only.

9.2 Trends across configurations

For Llama 3.1, mean lexical diversity increased from 0.017 in C1 to 0.395 in C4, while exact-match rate declined from 60% to 20%. For Qwen 2.5, mean lexical diversity increased from 0.118 in C1 to 0.383 in C4, while exact-match rate was 40% in C1 and 0% in C4. Qwen's diversity was not strictly monotonic: C3 (0.415) was higher than C4 (0.383).

The pattern is consistent with broader sampling often producing more varied wording and lower exact repeatability. However, because Temperature, Top-K and Top-P change together across configurations, the results cannot identify which individual parameter caused the differences.

9.3 Task-category observations

Model	Category	Rows	Mean diversity	Exact-match rate
llama3.1	Factual	8	0.056	87.5%
llama3.1	Reasoning	8	0.236	0.0%
llama3.1	General	4	0.427	25.0%
qwen2.5	Factual	8	0.276	25.0%
qwen2.5	Reasoning	8	0.147	12.5%
qwen2.5	General	4	0.634	0.0%

The general creative prompt showed the greatest lexical variation, particularly for Qwen 2.5. The factual prompts generally had lower variation for Llama 3.1 than the creative prompt. These category comparisons are descriptive and are based on only one general prompt and two prompts in each of the other categories.

9.4 Qualitative correctness observations

The two factual questions have straightforward reference answers: Canberra and Jane Austen. The logical implication prompt's correct conclusion is yes, because the subset/implication relationship is transitive. The bat-and-ball problem's correct ball price is $0.05. Manual inspection of the provided output samples shows that most responses state these expected answers, but at least one Llama 3.1 response at the highest setting contains an incomplete/incorrect calculation. Qwen's creative prompt also produced a response containing Chinese text in one run. These observations are examples, not a full accuracy score; a formal per-response rubric is needed before reporting accuracy percentages.

10. Streamlit Interface

The application allows a user to select llama3.1 or qwen2.5, adjust Temperature, Top-K and Top-P, choose one to five live runs and enter a prompt. Clicking 'Run Live Experiment' calls Ollama and displays each response in an expander. If experiment_results.csv exists in the working directory, the app displays the saved dataset. The supplied code displays a table rather than metric charts; plots can be added as a future improvement.

11. Limitations

- Only five prompts are included, limiting coverage and generalization.
- Three runs per configuration provide a small sample of output variability.
- Configurations change multiple parameters at once, so individual effects cannot be isolated.
- TF-IDF cosine distance measures lexical overlap, not meaning or factual quality.
- Exact string equality is strict and treats paraphrases as different responses.
- The experiment script does not automatically score factual accuracy or hallucination.
- The CSV stores three outputs joined into one field, which makes deeper per-run analysis less convenient.
- Model version, quantization, hardware and Ollama version should be documented for reproducibility.

12. Future Improvements

- Expand the prompt set and include more domains and difficulty levels.
- Vary one decoding parameter at a time while fixing the other two.
- Add reference answers and a human-scoring rubric for factual and reasoning accuracy.
- Store each response in a separate CSV row with a run ID and error/status fields.
- Add summary charts to Streamlit and provide downloadable CSV results.
- Record exact model tags and software versions in the README.
- Handle empty responses and TF-IDF vectorization errors explicitly.

13. Conclusion

The supplied results show that decoding configurations are associated with differences in lexical diversity and exact-match repeatability for both tested models. Across the 20 model/prompt/configuration rows per model, Llama 3.1 had lower average lexical diversity (0.161) and higher exact-match repeatability (40%) than Qwen 2.5 (0.296 diversity; 15% exact-match repeatability). The highest-temperature/broadest-sampling configuration produced more diversity than the lowest configuration for both models, though the pattern was not strictly monotonic for Qwen 2.5. These results are limited to a five-prompt benchmark and combined parameter settings. Factual accuracy and hallucination should be measured separately before making strong claims about correctness.

Appendix. Project Files

File	Purpose
app.py	Streamlit live-generation interface and saved-results table.
experiment.py	Batch experiment runner and metric calculations.
benchmark_prompts.json	Five benchmark prompts and their categories.
requirements.txt	Python dependencies: ollama, streamlit, pandas, numpy and scikit-learn.
experiment_results.csv	Collected results; 40 rows, each summarizing three generations.
Report (.docx)	Report with results and discussion based on the supplied CSV.